CHAPTER 9

# Exponential function and models continuous

**Chapter path**

**Objectives.** Use algebraic properties; differentiate; study where a function increases or decreases ; link the exponential and geometric sequences.
**Prerequisites.** Powers, geometric sequences, differentiation and sign.
**Yearly reference point.** 2·11 February: 8 a.m.

## Diagnostic check: six questions, 10 minutes

Answer without looking at your notes. Indicate the questions for which help is needed. This diagnostic test is not graded.

1. Simplify 2 $^{3} \times 2^{4}$.

2. Express 3 $^{-2}$.

3. Solve 2 $x + 1 = 5$.

4. Differentiate $x^2 + 2x$.

5. A decrease of 20% gives which multiplier?

6. For a geometric sequence $u_0 = 5, q = 2$, calculate $u_n$.

**After the diagnostic check.** Go over the flagged concepts with the teacher; redo a question similar, then explain the rule used.

**Oral work and logic.** Explain the difference between a continuous rate and a discrete multiplier; present an inequality proved via the minimum of a difference.

## Investigate before formalising

Compare the increments of $e^x$ and the values of the function around 0, with several steps positive. Carry out three Euler steps and explain the rule chosen.

**Organisation.** Individual work 5 min, discussion in pairs 5 min, whole-class sharing 5 min. Keep a record of an attempt, even an unfinished one. A conjecture must then be justified.

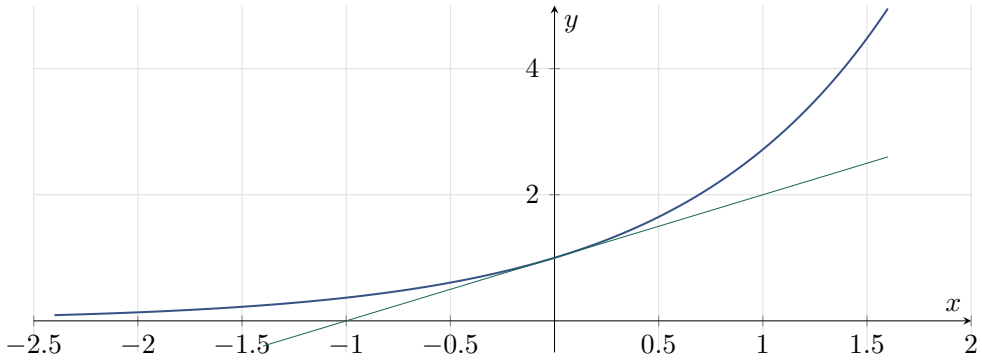
## 9.1 Defining a new function

**Definition and results**

The exponential function is the unique function f differentiable on $\mathbb{R}$ satisfying $f' = f$ and $f(0) = 1$. Its existence and uniqueness are assumed. It is denoted $\exp$, then $e^x$ with $e = \exp(1) \approx 2{,}71828$. The number e is a constant; x is the variable. The exponential is strictly positive.

**Status and justification.** Existence and uniqueness accepted, as in the syllabus. The positivity is deduced, after the algebraic properties assumed below, from $\exp(x) = \exp(\frac{x}{2})^2$ and $\exp(x)\exp(-x) = 1$.

**Worked example.** $\exp(0) = 1$, $\exp(1) = e$, $\exp'(0) = 1$. The tangent at 0 is therefore $y = 1 + x$. The notation $e^x$ extends the properties of powers to any real exponent.




$e^x$ (blue) and its tangent $1 + x$ at 0 (green).

**Watch out**

$e^0 = 1$, not 0. The function never takes the value 0, even for negative x.

**Immediate application.** What is the equation of the tangent to $e^x$ at 1?

## 9.2 Algebraic properties and comparisons

**Definition and results**

For all real numbers x, y, $e^{x+y} = e^x e^y$, $e^{-x} = \frac{1}{e^x}$, $e^{x-y} = \frac{e^x}{e^y}$. For integer n, $(e^x)^n = e^{nx}$. The function is strictly increasing on $\mathbb{R}$, therefore $e^u = e^v \iff u = v$ and $e^u < e^v \iff u < v$.

**Status and justification.** The addition formula is admitted in the common core; it can be explored further via uniqueness. Positivity and $\exp' = \exp > 0$ give strict monotonicity. This justifies equivalences for equations and inequalities.

**Worked example.** $e^{2x}e^{1-x} = e^{x+1}$. $e^{3x-2} = e^{x+4}$ is equivalent to 3 $x - 2 = x$ +4, so x=3. $e^{2x} \leq 1 = e^0$ is equivalent to $x \leq 0$.

**Watch out**

$e^{x+y}$ is a product, never in general $e^x + e^y$. We do not solve $e^x$ = 2 with a logarithm not studied in Première.

**Immediate application.** Simplify $\frac{e^{3x}}{e^{x-1}}$; solve $e^{x+2} > e^{2x}$.

### 9.3 Differentiate and study

**Definition and results**

For real a, $t \mapsto e^{at}$ is differentiable and its derivative is $ae^{at}$. For $e^{at+b} = e^b e^{at}$the derivative is $ae^{at+b}$. The product and quotient rules apply with derivatives already known. The sign of the exponential often simplifies the study of the sign of the derivative.

**Status and justification.** The differentiation formula for $e^{at}$ is admitted; the case with b follows from it by taking out the constant factor $e^b$. The general chain rule is not needed here.

**Worked example.** For $f(x) = (x+1)e^{-x}$, $f'(x) = e^{-x} - (x+1)e^{-x} = -xe^{-x}$. Since $e^{-x} > 0$, the derivative has the sign of $-x$ : global maximum 1 at 0.

**Watch out**

Do not forget the factor a or the minus sign in $e^{-x}$. The exponential factor is positive, not necessarily increasing.

**Immediate application.** Differentiate $g(t) = 3e^{-2t}$ and study how it varies.

## 9.4 From discrete to continuous

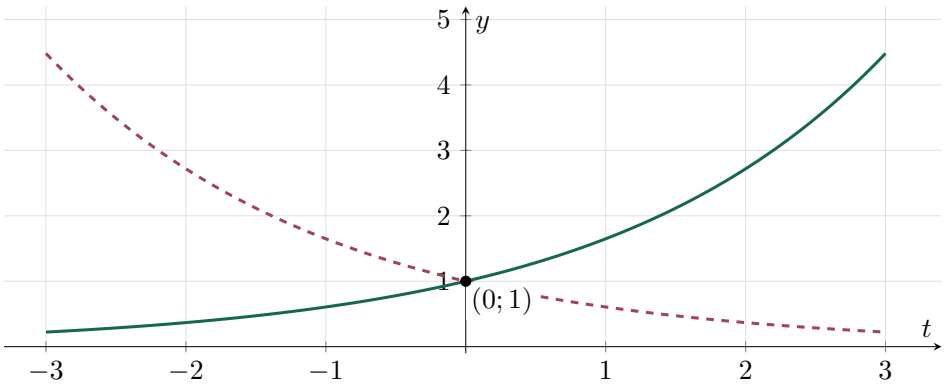
**Definition and results**

A model $Q(t) = Q_0 e^{kt}$, with $Q_0 > 0$, represents growth if $k > 0$, a decrease if $k < 0$, a constant if $k = 0$. For a fixed step h, $\dfrac{Q(t+h)}{Q(t)} = e^{kh}$. The values at times integers form a geometric sequence with common ratio $e^k$. The unit of k is the reciprocal of the unit of time.

**Status and justification.** The ratio comes from $e^{k(t+h)} = e^{kt}e^{kh}$. The variation is deduced from $Q'(t) = kQ_0e^{kt}$ and the positivity of the remaining factor.

**Worked example.** $Q(t) = 100e^{-0{,}2t}$, t in hours: $Q(0) = 100$, $\dfrac{Q(t+1)}{Q(t)} = e^{-0{,}2}$. A simulation can look for the first whole number of hours at which Q falls below 50; the exact equation is not solved by logarithm in this chapter.

ormalsize $e^{\frac{t}{2}}$ : increasing ormalsize $e^{-\frac{t}{2}}$ : decreasing




The curves corresponding to $k = \dfrac{1}{2}$ and $k = -\dfrac{1}{2}$ are symmetric about the vertical axis. The line styles also distinguish them without colour.

**Watch out**

The relative rate per unit of discrete time is $e^k - 1$, not exactly k; they are close only for small k.

**Immediate application.** For $Q(t) = 5e^{0{,}3t}$, express $Q(n)$ and $\dfrac{Q(t+2)}{Q(t)}$.

## 9.5 Build numerically and prove an inequality

**Definition and results**

To approximate the solution of $y' = y$, $y(0) = 1$ with one step $h$, Euler replaces the increment by $hy$ : $y_{k+1} = (1+h)y_k$. On [0; 1] with $h = \frac{1}{n}$, the approximation of $e$ is $\left(1+\frac{1}{n}\right)^n$. This construction is experimental. The inequality $e^x \geq 1 + x$ is obtained by studying a function.

**Status and justification.** For $g(x) = e^x - 1 - x$, $g'(x) = e^x - 1$, negative if x<0 and positive if x>0. The minimum is $g(0) = 0$, which proves the inequality; equality only at 0. Convergence of Euler is not proved here.

**Worked example.** With n=2, Euler's method gives $y_0 = 1, y_1 = 1{,}5, y_2 = 2{,}25$. With n=10, $(1\ \ ,1)^{10} \approx 2{,}59374$. We compare with e; we do not confuse an approximate value with the definition of the function.

**Watch out**

A graph showing a curve above a line does not prove the inequality for all the real numbers.

**Immediate application.** Show that $e^{2x} \geq 1 + 2x$ for all real x.

## Exercise path

Start with exercises 1 and 2. Then choose exercise 3 or 4 without help. Exercises 5 to 7 are a reserve for transfer, investigation or programming: the teacher chooses one. Exercise 8 is revisited later.

### Exercise 9.1 · Transform

For every real number $x$, we consider the expressions

$$A = e^x e^{2-x}, \qquad B = \frac{(e^{2x})^3}{e^x}, \qquad C = e^{-x}(e^x + 1).$$

1. Simplify $A$ and $B$ justifying each grouping of exponents.
2. Expand then simplify $C$. Explain why a sum does not simplify in the same way than a product.

### Exercise 9.2 · Equations and inequalities

Consider an equation and an inequality in a real unknown $x$ :

$$e^{2x-1} = e^{x+3} \quad \text{and} \quad e^{-3x} \geq e^{x-2}.$$

1. Solve each one in $\mathbb{R}$, citing the property of the exponential function that allows the comparison of the exponents.
2. When solving the inequality, explain at which step the direction of the inequality changes. Give the solution set as an interval.

### Exercise 9.3 · Product and maximum

The function $f$ is defined on [0; 4] by $f(x) = xe^{-x}$.

1. Calculate $f\prime(x)$, then study its sign on the interval.
2. Draw the variation table of $f$. Deduce its maximum and the x-value where it is attained.
3. Check that the values at the endpoints do not give a larger maximum. Keep the exact values.

### Exercise 9.4 · A modelled population

The following two parts are independent.

**Part A - A population.** A population is modelled by $P(t) = 200e^{0,1t}$, where $t \geq 0$ is the time elapsed in years. $P(t)$ is an expected count, which need not be an integer.

1. Determine the initial count, then the multiplier between times $t$ and $t + 1$. In deduce the exact annual rate and its percentage rounded to the nearest hundredth.
2. Explain why this model cannot necessarily describe a real population indefinitely.
3. **Part B - Two curves.** In the same coordinate system, sketch the graphs of $t \mapsto e^{\frac{t}{2}}$ and $t \mapsto e^{-\frac{t}{2}}$. State their variations, their common point and their symmetry. Clearly identify each curve.

### Exercise 9.5 · Search for a threshold

The amount of a substance, in milligrams, is modelled at time $t \geq 0$ by $Q(t) = 80e^{-\frac{t}{2}}$, where $t$ is expressed in hours. Checks take place at whole-number times $n = 0, 1, 2, \ldots$. We seek the first check for which the quantity is strictly less than 10 mg.

1. Write an algorithm to find this first integer. Include a bound on the number of iterations and specify what the program returns if the bound is reached.
2. Run the algorithm, then check numerically the quantity at the index found and at the previous index.
3. Justify, using the decrease of $Q$, that this double check shows that it is indeed the first index. Do not use logarithms.

### Exercise 9.6 · A symmetry

Consider the function $f$ defined on $\mathbb{R}$ by $f(x) = e^x + e^{-x}$.

1. Study the parity of $f$ and interpret the result for its curve.
2. Prove that $f(x) \geq 2$ for every real $x$, choosing a method. State the equality case.
3. Suggest checking the result by a second method, using a square or a study of variations.

### Exercise 9.7 · Uniqueness under an assumption

**Extension.** Assume that the exponential function has already been constructed. Let $f$ a function differentiable on $\mathbb{R}$ which satisfies $f'(x) = f(x)$ for every real $x$ and $f(0) = 1$. Set $g(x) = f(x)e^{-x}$.

1. Calculate $g'$ and explain why $g$ is constant on $\mathbb{R}$.

2. Determine this constant and then deduce from it the expression of $f$.

3. Does this proof establish the existence or the uniqueness of the required function? Justify.

### Exercise 9.8 · Delayed review

Define the sequence $(u_n)$ by $u_n = e^{2n+1}$ for every natural number $n$.

1. Prove that this sequence is geometric, then give its first term and common ratio.

2. Calculate $u_0 + u_1 + u_2$ in exact form. Compare the direct calculation with using a formula of a sum.

**Independent work.** Do only the task assigned at the end of the session, in the scheme of work Date: 10·20 min maximum. The exercises depend on the concepts already studied; some questions are therefore postponed. Revisit exercise 8 one week after the chapter (10 min). If the planned time is exceeded, note the difficulty and bring your attempts; do not use up the whole reserve.

## To be done in the CQFD textbook

These exercises are done in your textbook. The scheme of work gives the session; they replace the personal work planned that day (10 to 20 minutes in total).

**CQFD, exercise 18 p. 187, questions a, b and c.** Simplify the products and quotients, citing the property of exponentials used.

**CQFD, exercise 30 p. 187, questions a, b and c.** Solve the three inequalities and write the solution sets in the form of intervals.

## MCQ 1: understand and choose

Five teaching questions with a single answer. Choose one answer per question; then explain one distractor. This chapter format does not reproduce the official format of the EAM.

1. $e^0 =$

   A. $0$
   B. $1$
   C. $e$
   D. $-1$

2. $e^x e^y =$

   A. $e^{x+y}$
   B. $e^{xy}$
   C. $e^x + e^y$
   D. $e^{x-y}$

3. Derivative of $e^{-2x}$ :

   A. $e^{-2x}$
   B. $2e^{-2x}$
   C. $-e^{-2x}$
   D. $-2e^{-2x}$

4. $e^x$ = 1 is equivalent to

   A. $x = 1$
   B. $x = e$
   C. $x = 0$
   D. no solution

5. The sign of $e^{-x}$ is

   A. negative if x>0
   B. Always strictly positive
   C. zero at 0
   D. Variable

## MCQ 2: understand and choose

Five teaching questions with a single answer. Choose one answer per question; then explain one distractor. This chapter format does not reproduce the official format of the EAM.

1. $e^{2x} < e^3$ is equivalent to

   A. $x < 3$
   B. $x > \frac{3}{2}$
   C. $x < \frac{3}{2}$
   D. $x < \sqrt{3}$

2. $u_n = e^{3n}$ has common difference/ratio

   A. $e^3$
   B. $3$

C. $e$

D. $3e$

3. Tangent at $e^x$ at 0:

A. $y = x$

B. $y = ex$

C. $y = 1$

D. $y = x + 1$

4. Derivative of $xe^x$ :

A. $e^x$

B. $(x + 1)e^x$

C. $xe^x$

D. $xe^{x-1}$

5. $e^x \geq 1 + x$ has equality

A. for every x

B. at 1 only

C. at 0 only

D. Never

## True or false? Justify.

1. $e^{2x} = 2e^x$ for all x.

2. The function $t \mapsto e^{-3t}$ is decreasing.

3. A continuous rate k=0.1 corresponds exactly to 10% increase in one unit of time.

**Personal review**

I keep the power rules; I know positivity; I differentiate with the factor; I separate
model, exact calculation and simulation.
For each objective, indicate: mastered without help; achieved with help; needs more work. Write an example and
a mistake to avoid.

# Solutions and teaching notes, chapter 9

**Teacher use.** The following answers concern the tasks not solved in the lesson. Do first make the approach explicit; provide graduated help before the solution. The durations are estimates to be calibrated in class.

## Diagnostic and remediation

1. $2^7 = 128$. *Action:* Reconstruct the product of seven factors.
2. $\frac{1}{9}$. *Action:* Go back to the reciprocal power.
3. $x = 2$. *Action:* Isolate the unknown while maintaining equivalence.
4. $2x + 2$. *Action:* Apply the sum of the derivatives.
5. $0{,}8$. *Action:* Subtract the rate from 1.
6. $5 \times 2^n$. *Action:* Count the n multiplications.

## Investigation

The condition $f\prime = f$ motivates ?$y \approx hy$. With h= $\frac{1}{4}$, the values are 1 ; 1.25 ; 1.5625 ; 1.953125. This construction approximates the function and does not replace its characterisation.

## Direct applications

1. $y = e + e(x - 1) = ex$.
2. $e^{2x+1}$ ; $x + 2 > 2x$, therefore $x < 2$.
3. $g\prime(t) = -6e^{-2t} < 0$ on $\mathbb{R}$ : strictly decreasing.
4. $u_n = 5(e^{0,3})^n$ ; the ratio over two units is $e^{0,6}$.
5. Apply the proved inequality with argument 2 $x$ ; equality only at x=0.

## Exercises: methods and answers

**Exercise 9.1 · Transform.** $e^2$ ; $e^{5x}$ ; $1 + e^{-x}$. For the last expression, distribute and simplify the products.

**Exercise 9.2 · Equations and inequalities.** Since it is strictly increasing: 2 $x - 1 = x$ + 3, i.e. x=4. Then $-3x \geq x - 2$, i.e. $x \leq \frac{1}{2}$.

**Exercise 9.3 · Product and maximum.** $f\prime = (1 - x)e^{-x}$ ; increasing up to 1, then decreasing. Maximum $f(1) = \frac{1}{e}$. Values at the endpoints 0 and 4 $e^{-4}$ lower.

**Exercise 9.4 · A modelled population.** 200 ; factor $e^{0,1}$ ; rate $e^{0,1} - 1 \approx 10{,}52\%$. The resources and the capacity of the environment are not taken into account; it is a model over a period of time bounded. The two required curves pass through (0;1); the first increases, the second decreases. They are symmetric about the vertical axis; see the figure in the section From discrete to continuous.

**Exercise 9.5 · Search for a threshold.** Test n=0,1,... in turn with a limit on the number of iterations. $Q(4) \approx 10{,}8268 \geq 10$, $Q(5) \approx 6{,}5668 < 10$ ; decrease proved, so the first integer is 5.

**Exercise 9.6 · A symmetry.** $f(-x) = f(x)$ : even. $f' = e^x - e^{-x}$ has the sign of x, so minimum 2 at 0. Another proof: $f(x) - 2 = (e^{\frac{x}{2}} - e^{-\frac{x}{2}})^2$.

**Exercise 9.7 · Uniqueness under an assumption.** $g' = f'e^{-x} - fe^{-x}$ = 0. On $\mathbb{R}$, g is constant, and g(0)=1. Thus $f(x) = e^x$. This proof of uniqueness uses an exponential already constructed; it does not does not prove existence.

**Exercise 9.8 · Delayed review.** $\frac{u_{n+1}}{u_n} = e^2$ ; first term e. Sum $e(1+e^2+e^4) = \frac{e(1-e^6)}{1-e^2}$. The direct sum is also effective here.

**CQFD, exercise 18 p. 187, questions a, b and c.** a : $e^{3x}$. b : $e^2$, since $e^{x+2}$ is divided by $e^x$, always non-zero. c: $e^{x+1}$. Do not confuse the number e with the variable x.

**CQFD, exercise 30 p. 187, questions a, b and c.** a : $x \leq 0$. b : $2x > 1$, therefore $x > \frac{1}{2}$. c : $-3x \geq 2x + 1$, therefore $x \leq -\frac{1}{5}$. Since the exponential is strictly increasing, the direction is kept; dividing by $-5$ reverses it in c.

## MCQ: answers and reasons

**MCQ 1.**

1. B. Initial condition.
2. A. Addition property.
3. D. Factor -2.
4. C. Strictly increasing and $e^0 = 1$.
5. B. Positivity of the exponential.

**MCQ 2.**

1. C. Compare the exponents.
2. A. Ratio of consecutive terms.
3. D. Value 1 and slope 1.
4. B. Product rule.
5. C. Minimum of the difference.

## True or false

1. False: at 0, 1 is different from 2.
2. True: its derivative $-3e^{-3t}$ is negative.
3. False: the discrete rate is $e^{0,1} - 1$, i.e. about 10.52%.

**Differentiated teaching.** If a pupil is stuck on a technical point, suggest only the first step (domain, factor or reference sample space), then remove the help. For an independent student, ask for a second method and the handling of a boundary case. Orally, allow an initial rephrasing in the language mastered, then require a precise mathematical sentence in French.